END TERM EXAMINATION
(CBCS)(SUBJECTIVE TYPE)
Course Name: B.Tech (IT) , Semester: 4th
(MAY, 2025)

Subject Code: BIT 204 | **Subject: Object Oriented Programming**

Time :3 Hours | **Maximum Marks :60**

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(5*4=20)	CO mapping
1	Explain -i)Derived and User defined data types ii) copy constructor		CO1
2	What is Class ? How function overloading can be achieved in C++?		CO2
3	Explain : i) reading data from a file in C++ ii)Java Virtual Machine		CO3
4	Explain in Java : i)how to use and create packages ii) default and public access specifiers in java		CO4
	UNIT-I		
Q2	a) Write a program in C++ that takes two numbers in the range from 1 to 50 from the user. It then compares these numbers against a single dimension array of five integer elements ranging in value from 1 to 50. The program displays the message NUMBERFOUND if the two inputted values are found in the array element. b) Using program explain use of inline functions in C++	(6,4)	CO1
Q3	What are references,pointers and objects in C++?Create two classes Rectangle(private:int dimension1,int dimension2,int area)and Triangle(private:int dimension1,int dimension2,int area).Create appropriate constructors and function in both classes for initializing and display the details.Use friend function to compare area of Rectangle and Triangle.	(10)	CO1
	UNIT-II		
Q4	What are virtual functions in C++?Create an abstract class in C++ called UniversityEmployee(private:int Basic,int TA,int DA, int HRA) and has pure virtual function called int salary() to calculate the salary.Create two classes namely TeachingEmployee and NonTeachingEmployee which are inheriting the class UniversityEmployee. The salary of any employee is the sum of Basic(B),Travelling allowance(TA),Dearness Allowance(DA) and HRA.But theTeaching employees get an extra allowance namely Special allowance(SA) of Rs. 20,000/-.In main function create objects of these classes and show the salary of employees.	(10)	CO2
Q5	What is compile time polymorphism?Create a Class in C++ which provides representation for the concept of Time(hours,minute,seconds). Using operator overloading, perform the operations of unary increment of a given Time. Implement operator overloading through friend function and member function.	(10)	CO2
	UNIT-III		
Q6	What is random file access in C++?How do you open a file in C++?Create a program in C++ to enter the details of the students (string name,int roll_no and int total_marks)and check if the total_marks of student is greater than 50 then save the details of student in Pass.txt else in Fail.txt.	(10)	CO3
Q7	Explain i) exceptional handling mechanisms in C++ and create a program to throw an exception division by zero ii)role of scope resolution operator and using keyword in namespace.	(10)	CO3
	UNIT-IV		
Q8	Explain using program a) interface in java b) Multithreading	(10)	CO4
Q9	What is inheritance in Java? Explain using the program its types and how to use super class constructor, variables and its methods in subclass.	(10)	CO4